

Introduction to Neural Networks

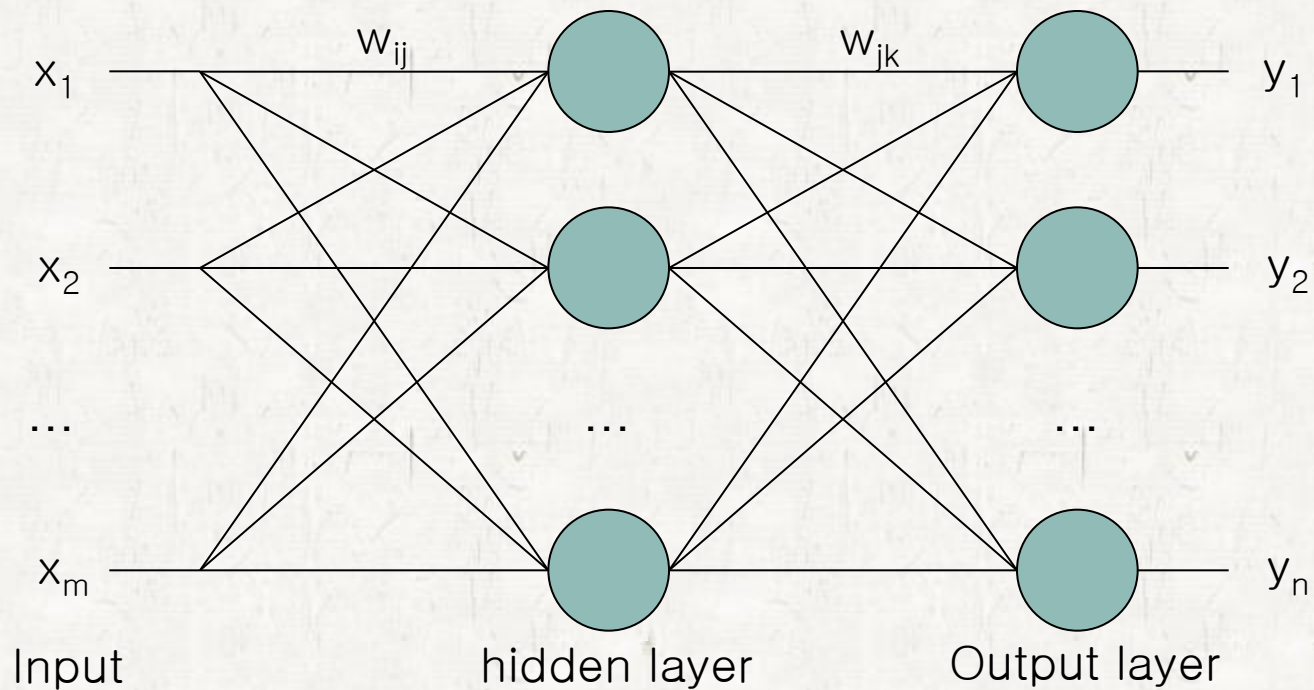
Introduction

Deep Learning = Deep Neural Network

- Deep? 그럼 Shallow Neural Network도 있겠네?
- Deep Neural Network 도 있고 Shallow Neural Network 도 있으면,
- 그럼, Neural Network은 뭐야?
- 왜, Neural 이야?

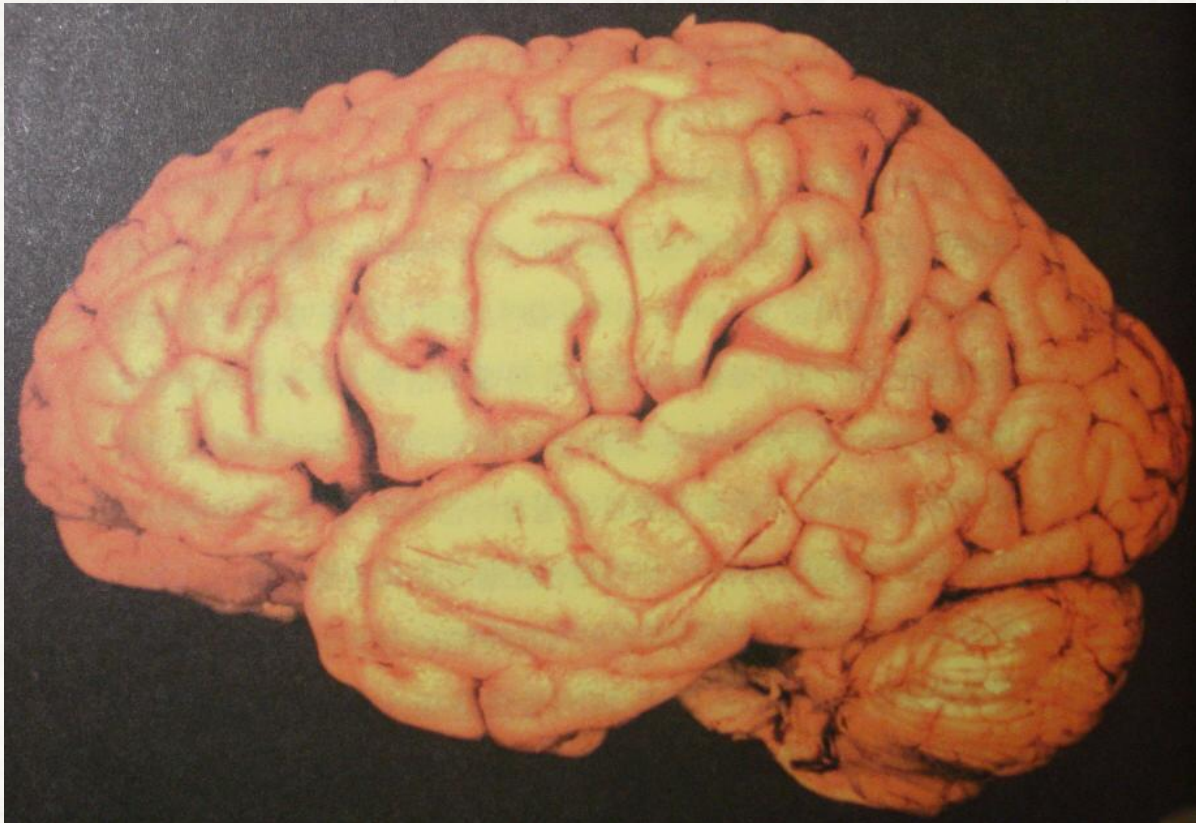
Introduction

● Neural Network



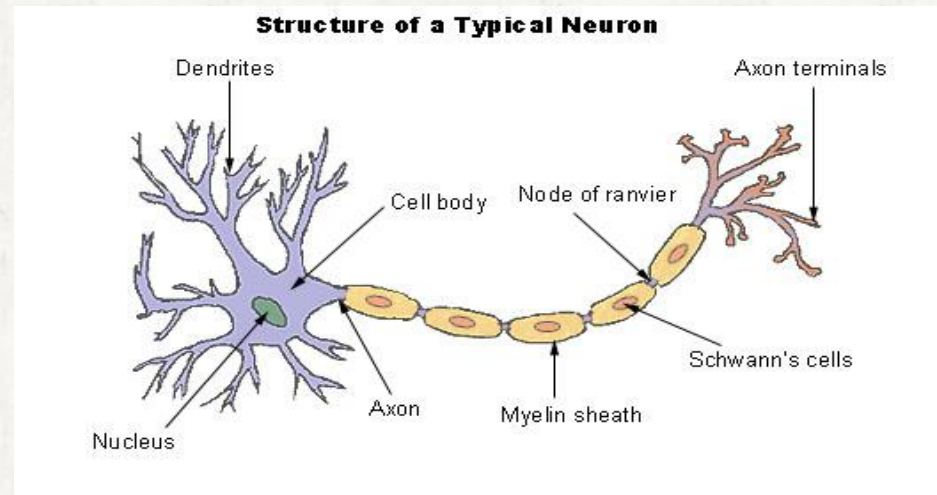
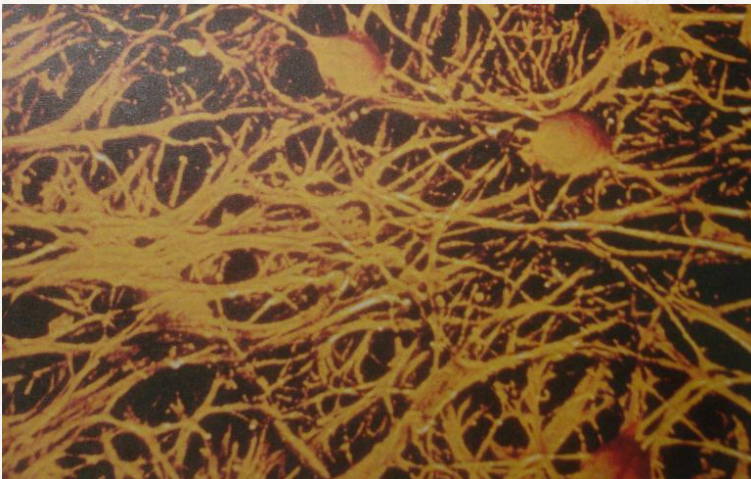
Introduction

- Brain
 - There are about 10^{11} neurons (brain cells)



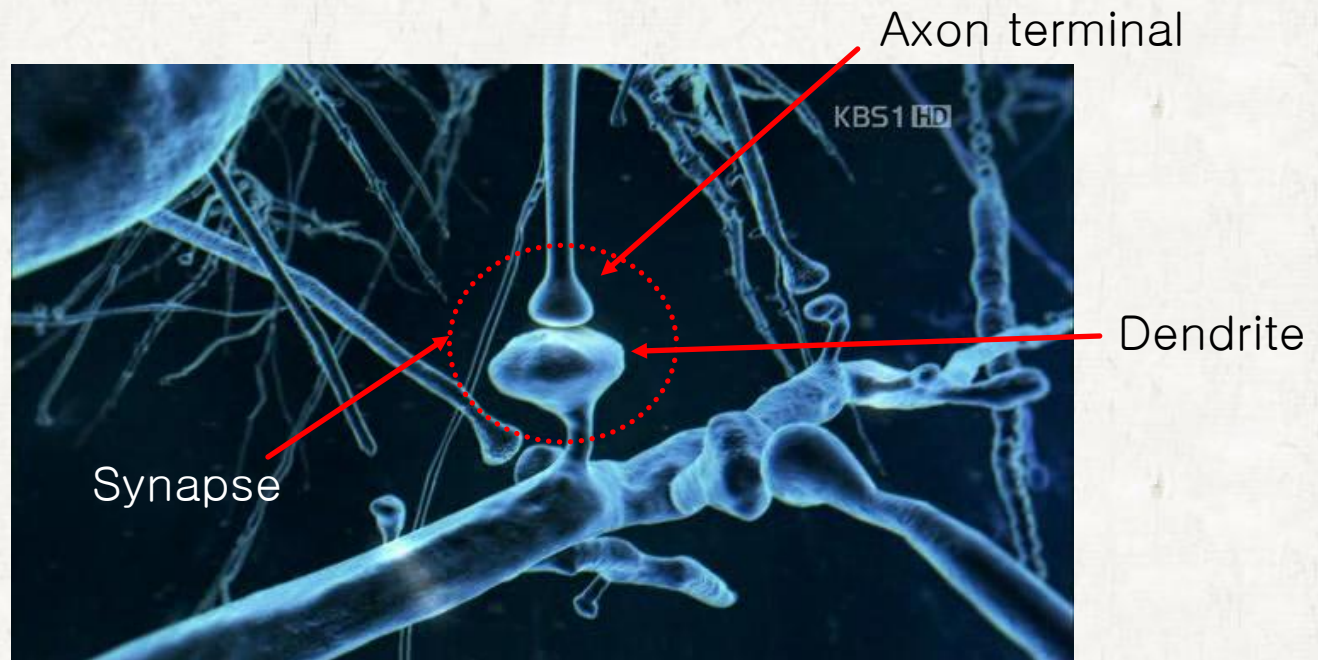
Introduction

- Neurons
 - Shape: Cell body, Dendrite, Axon
 - Every neuron connects to 10^3 to 10^4 other neurons
 - A brain is a network of neurons



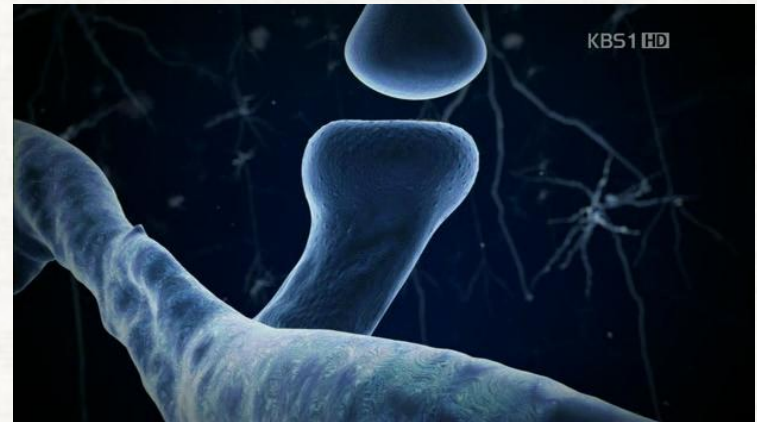
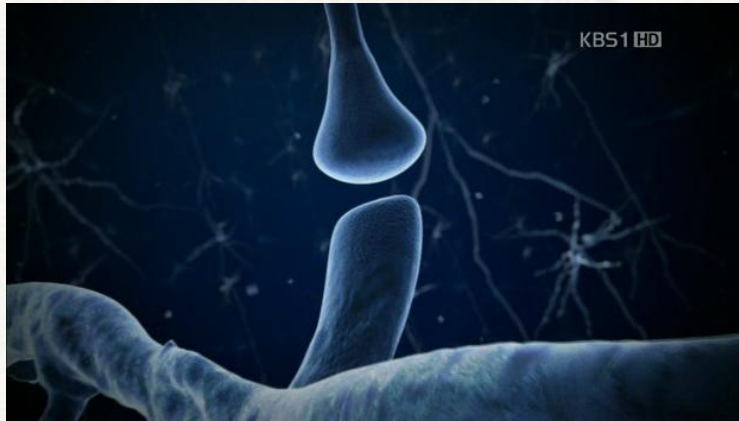
Introduction

- Connection between neuron
 - Synapse : Connection spot
 - Axon terminal : release neurotransmitter
 - Dendrite : receive neurotransmitter



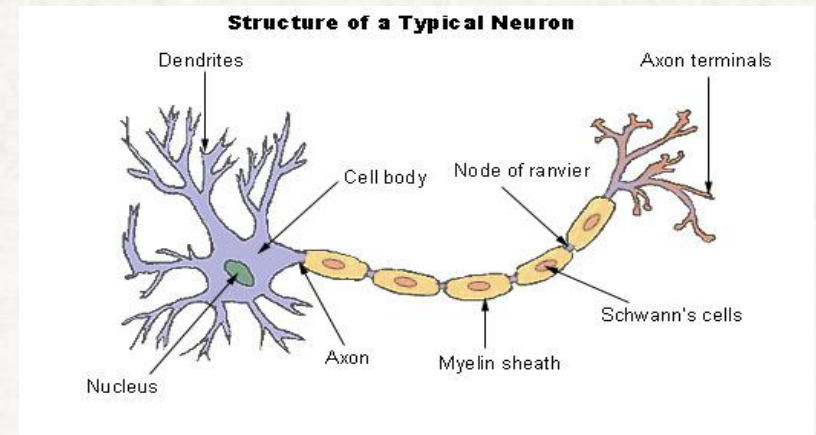
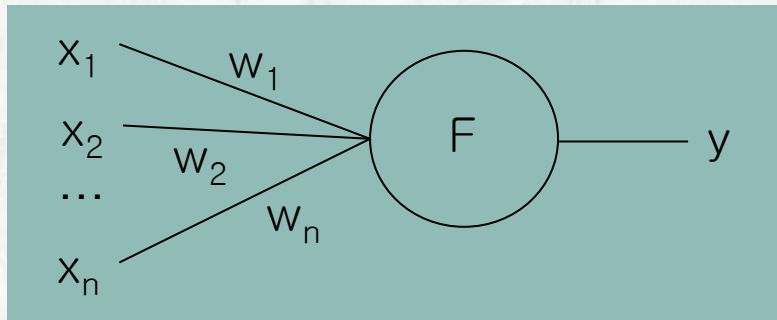
Introduction

- Connection between neuron
 - Every connection does not have the same effect
 - Each connection has different strength
 - The more receptor a dendrite has (mushroom shape), the better it receives neurotransmitter



Simple Mathematical Model

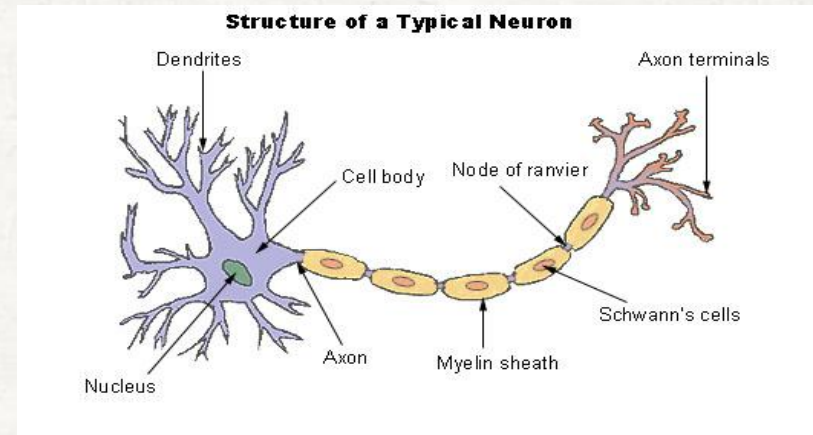
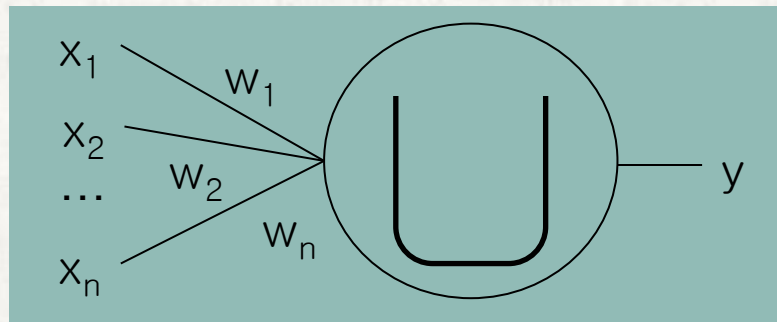
● Simple representation of neurons



- x : dendrites (input)
- w : amount of receptors in each dendrite (connection strength)
- F : cell body
- y : axon (output)

Simple Mathematical Model

- Abstract Behavior of Neuron
 - Neuron is a cell not a computer
 - When signals come in, it just absorbs them.



- When signals come in, a neuron collects them in the bucket
 - If the amount of signal reaches the threshold, it allows the signals to flow out
 - Otherwise, it does not output

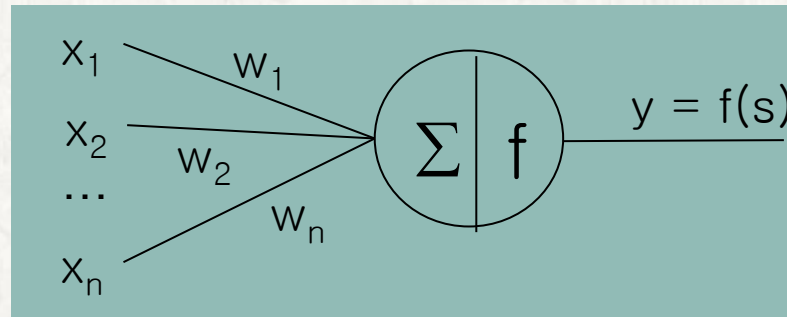
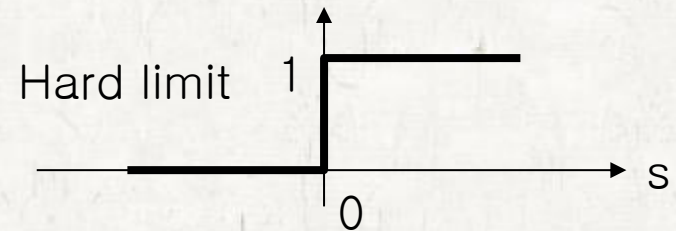
Simple Mathematical Model

- Simple mathematical model of neurons—con'd
 - First function: Weighted summation of inputs

$$s = x_1w_1 + x_2w_2 + \dots + x_nw_n$$

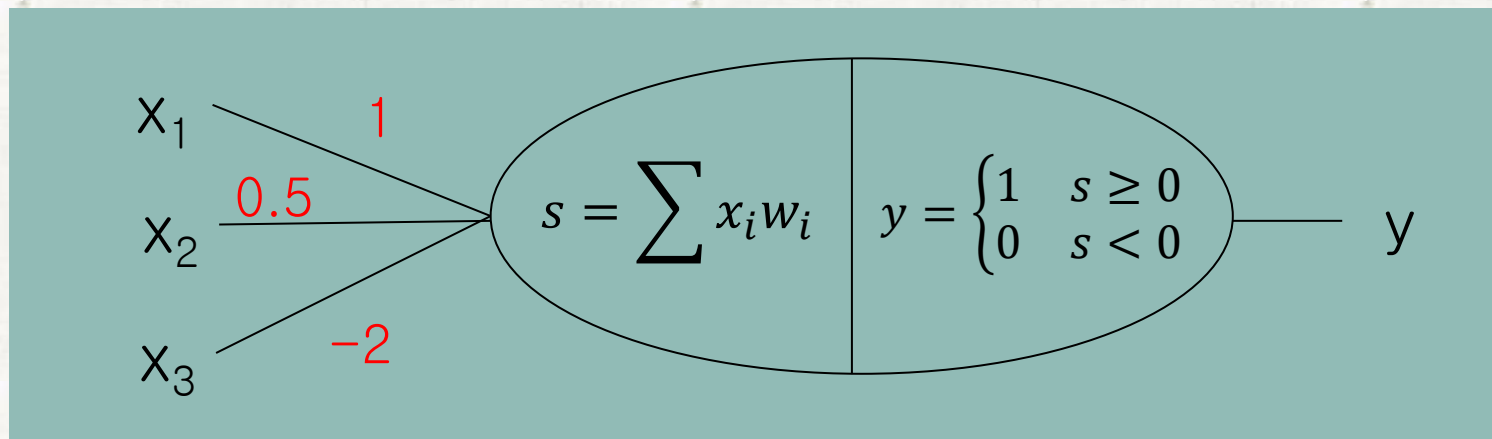
- Second function: Non-linear threshold

$$y = \begin{cases} 1 & s \geq 0 \\ 0 & s < 0 \end{cases}$$



Simple Mathematical Model

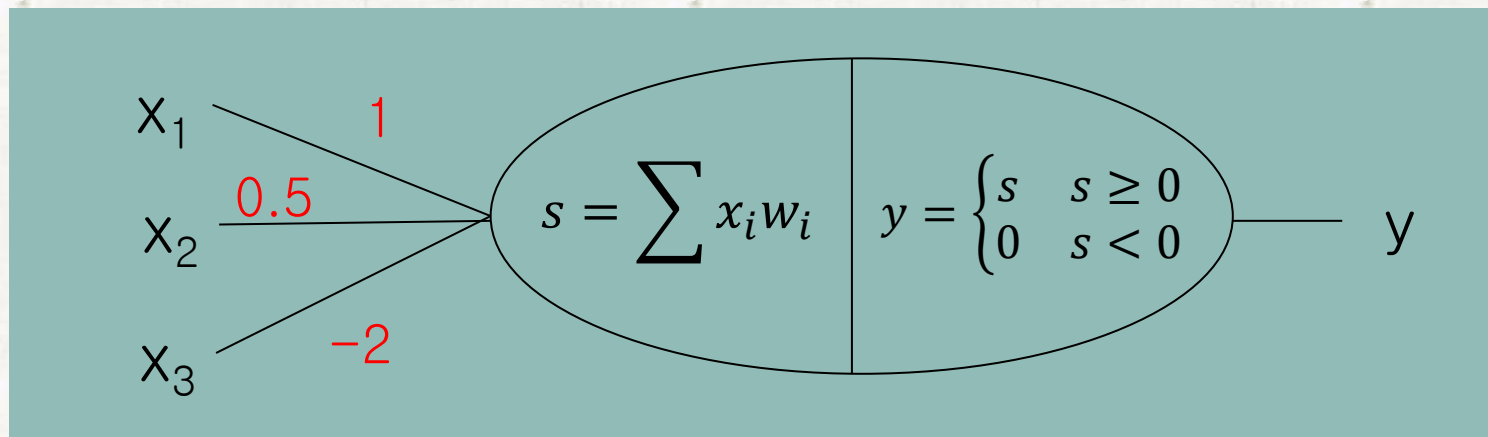
- Simple mathematical model of neurons—con'd



x_1	x_2	x_3	Σ	y
0	0	1	$0 \cdot 1 + 0 \cdot 0.5 + 1 \cdot (-2) = -2$	0
-0.5	3	1	$(-0.5) \cdot 1 + 3 \cdot 0.5 + 1 \cdot (-2) = -1$	0
0.5	1	-0.5	$0.5 \cdot 1 + 1 \cdot 0.5 + (-0.5) \cdot (-2) = 2$	1
2	1	0.6	$2 \cdot 1 + 1 \cdot 0.5 + 0.6 \cdot (-2) = 0.3$	1

Simple Mathematical Model

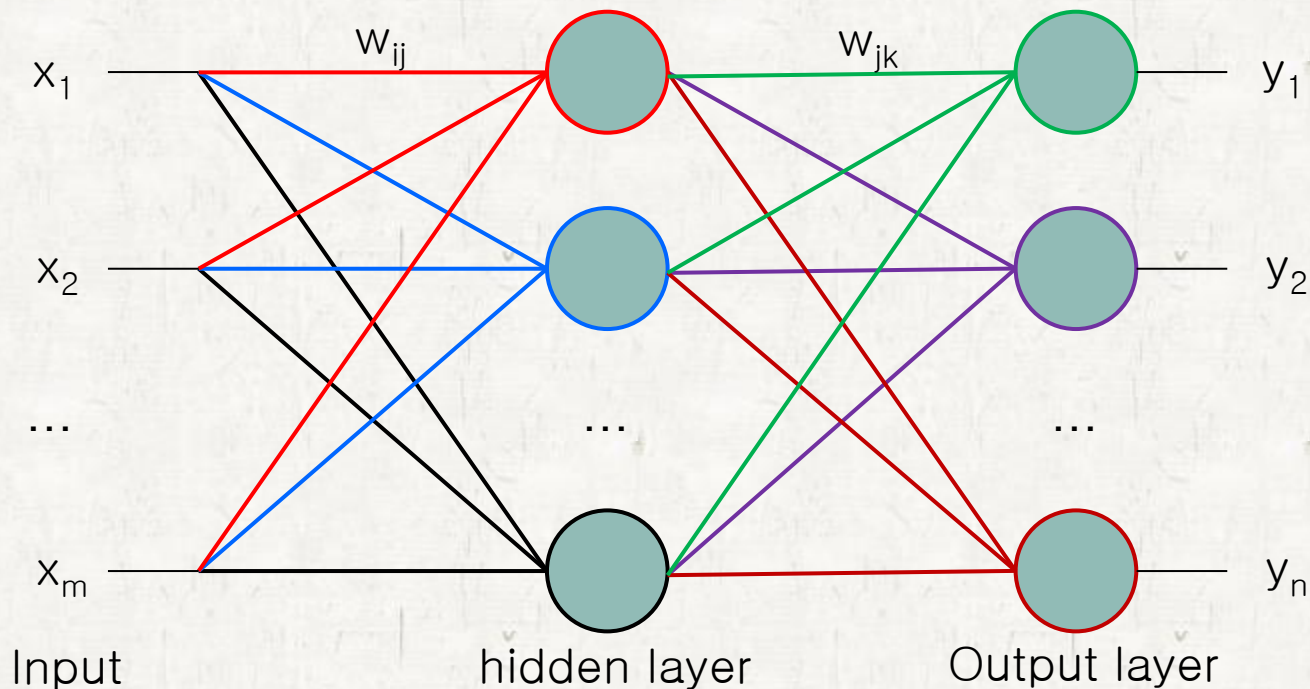
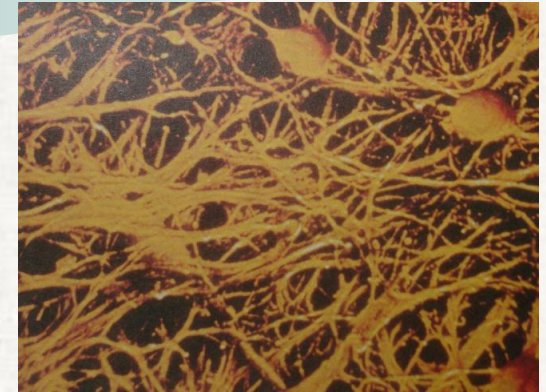
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2	1	0.6	$2 \cdot 1 + 1 \cdot 0.5 + 0.6 \cdot (-2) = 0.3$	0.3

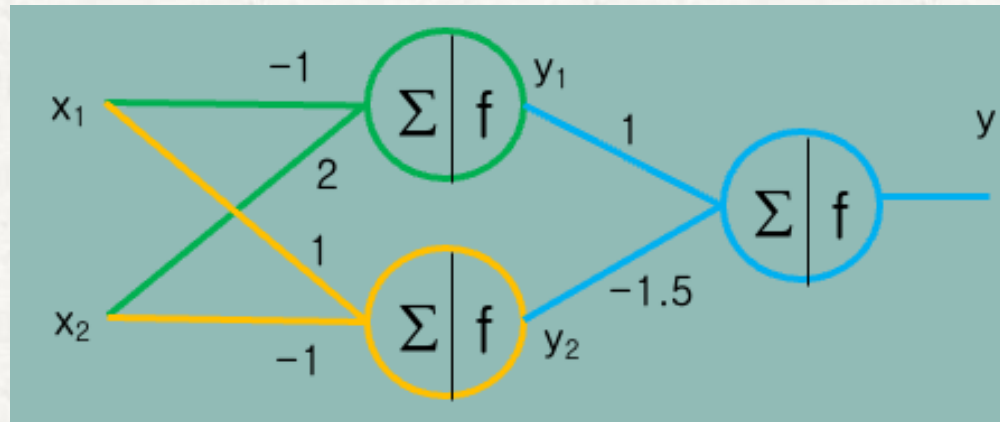
Neural Network

- Structure of Neural Network
 - Brain is a network of neurons
 - So, let's simply connect artificial neurons and call it artificial neural network



Neural Network

- What a neural network can do– con'd



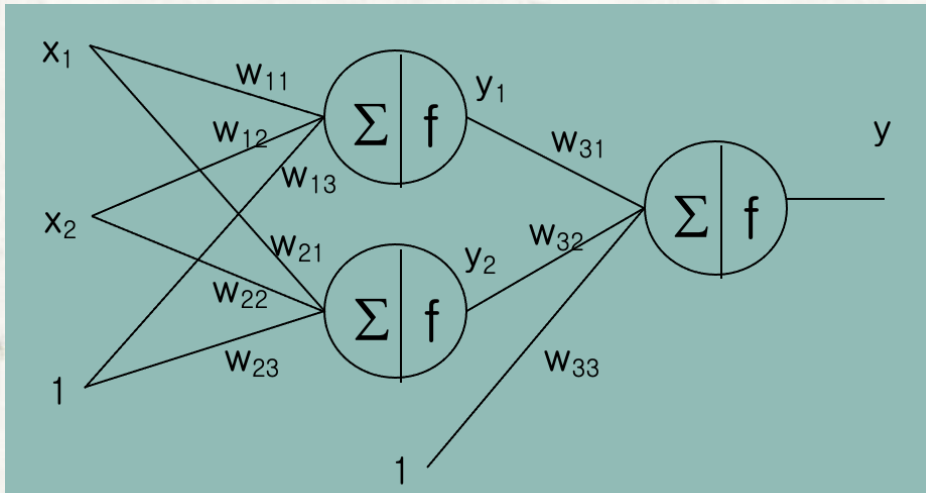
x_1	x_2	s_1	y_1
0	1	2	1
2.5	1	-0.5	0
-1	0.5	1	1
2	1.5	1	1

x_1	x_2	s_2	y_2
0	1	-1	0
2.5	1	1.5	1
-1	0.5	-0.5	0
2	1.5	0.5	1

y_1	y_2	s	y
1	0	1	1
0	1	-1.5	0
1	0	1	1
1	1	-0.5	0

What a Neural Network Can Do? (2)

- What a neural network can do– con'd



$$w_{11}=1.0, w_{12}=1.0, w_{13}=-1.5$$

$$w_{21}=1.0, w_{22}=1.0, w_{23}=-0.5$$

$$w_{31}=-1.0, w_{32}=1.0, w_{33}=-0.5$$

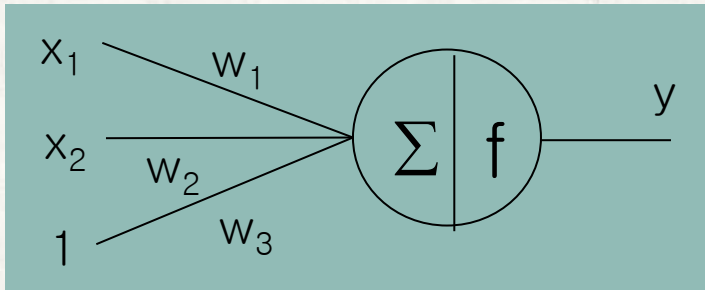
x_1	x_2	Σ	y_1
0	0	-1.5	0
0	1	-0.5	0
1	0	-0.5	0
1	1	0.5	1

x_1	x_2	Σ	y_2
0	0	-0.5	0
0	1	0.5	1
1	0	0.5	1
1	1	1.5	1

y_1	y_2	Σ	y
0	0	-0.5	0
0	1	0.5	1
0	1	0.5	1
1	1	-0.5	0

What a Perceptron Can Do?

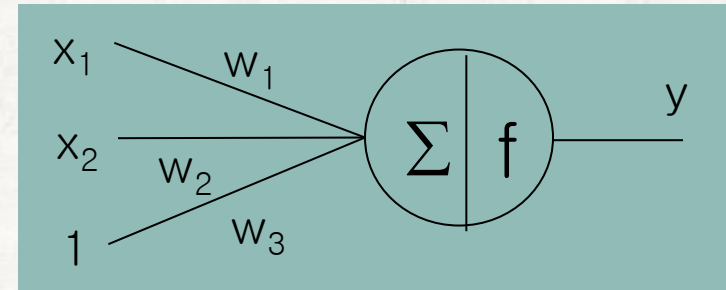
- What a perceptron can do
 - And operation



$w_1=1.0$, $w_2=1.0$, $w_3=-1.5$

x_1	x_2	Σ	y
0	0	-1.5	0
0	1	-0.5	0
1	0	-0.5	0
1	1	0.5	1

OR operation

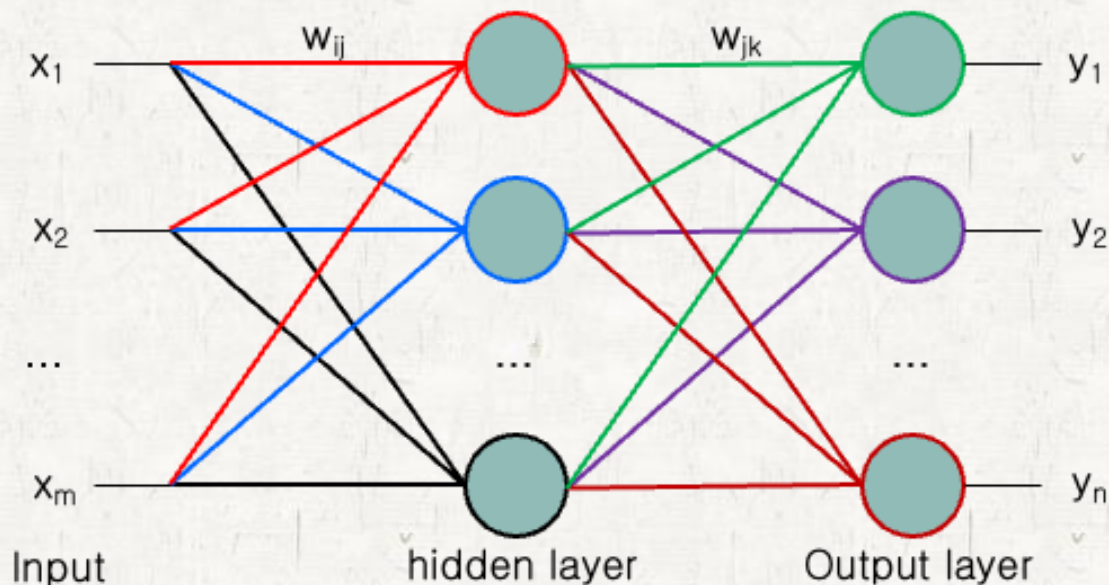


$w_1=1.0$, $w_2=1.0$, $w_3=-0.5$

x_1	x_2	Σ	y
0	0	-0.5	0
0	1	0.5	1
1	0	0.5	1
1	1	1.5	1

What a Neural Network Can Do?

- Simple mathematical model of brains—con'd



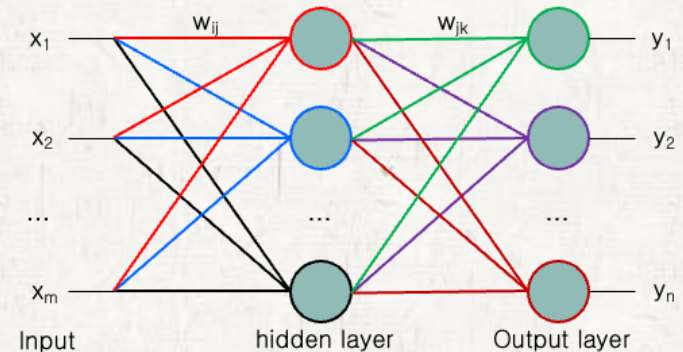
- How stupid it is!!
- What can it do?
 - Everything a computer can do!!

What a Neural Network Can Do?

- Simple mathematical model of brains—con'd



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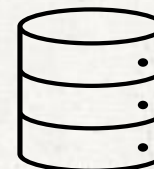
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Connection weights
in NN

Programming
by Human



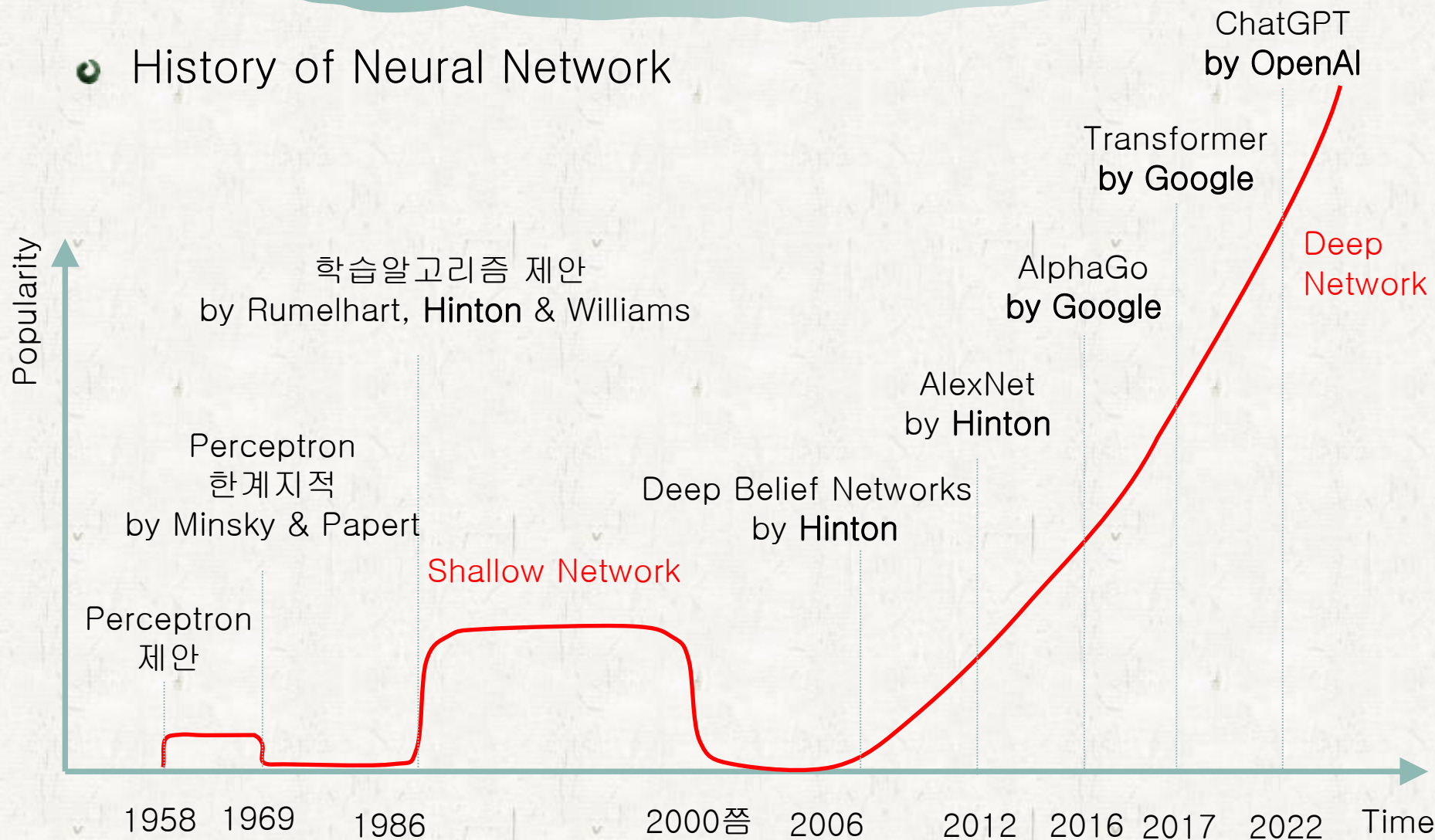
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Training with Data

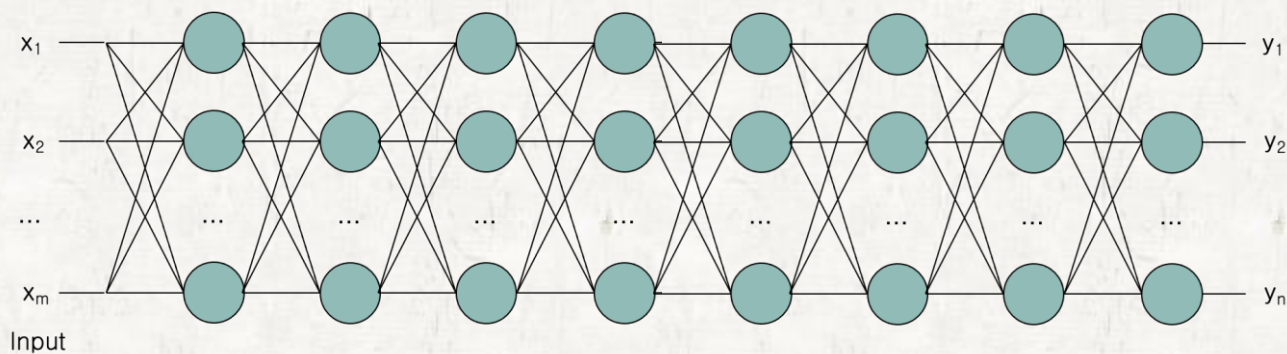
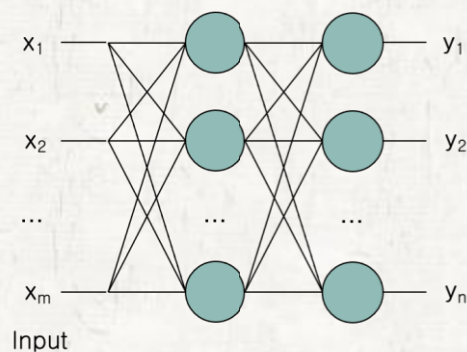
What a Neural Network Can Do?

History of Neural Network



Neural Network

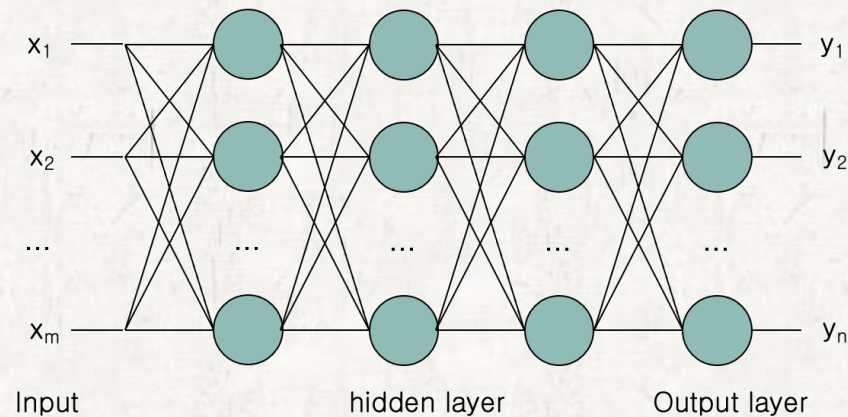
Why deep?



Neural Network Design

● Neural Network Structure

- How many inputs?
- How many outputs?
- How many hidden layers?
- How many nodes in hidden layers?



Neural Network Design

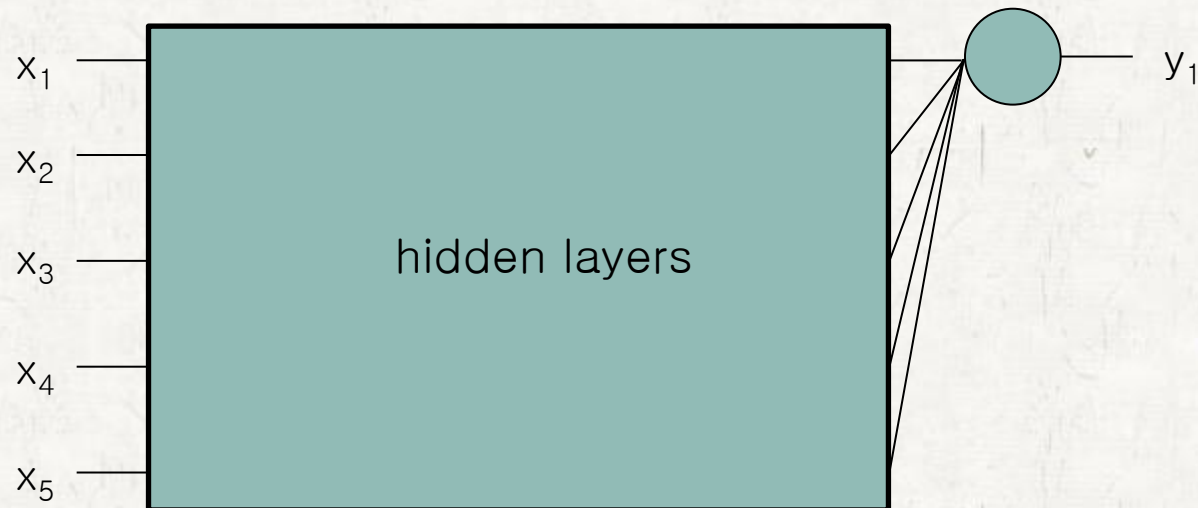
- How many inputs? How many outputs?

- Stock Index Prediction

(2500, 2550, 2530, 2540, 2550) $\rightarrow$ 2600

(2400, 2410, 2420, 2430, 2440) $\rightarrow$ 2450

(2470, 2460, 2450, 2470, 2480) $\rightarrow$ 2470



Neural Network Design

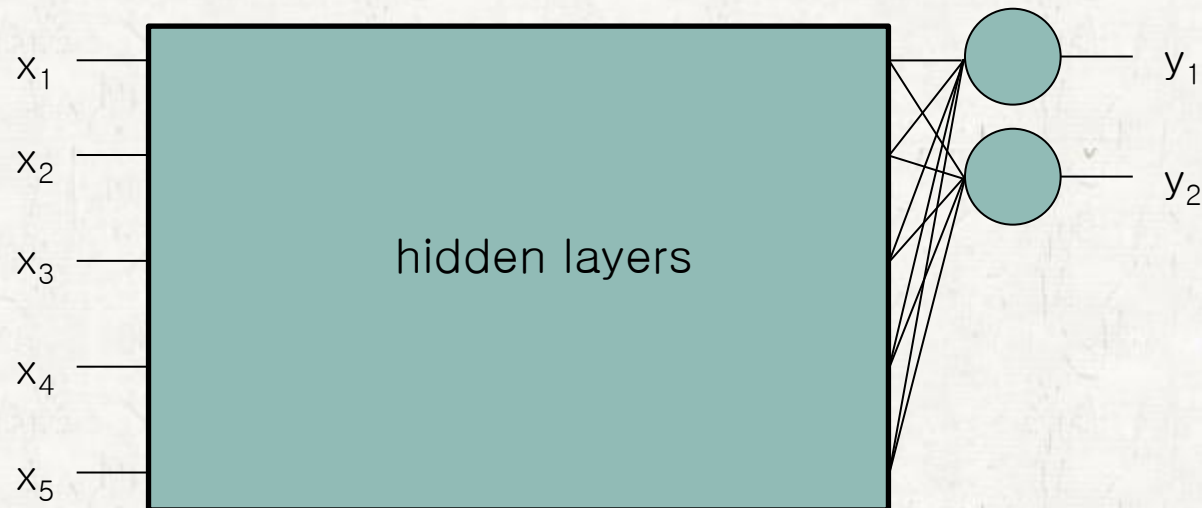
- How many inputs? How many outputs?

- Stock Index Prediction

(2500,2550, 2530, 2540, 2550) $\rightarrow$ (2600,2580)

(2400,2410, 2420, 2430, 2440) $\rightarrow$ (2450,2460)

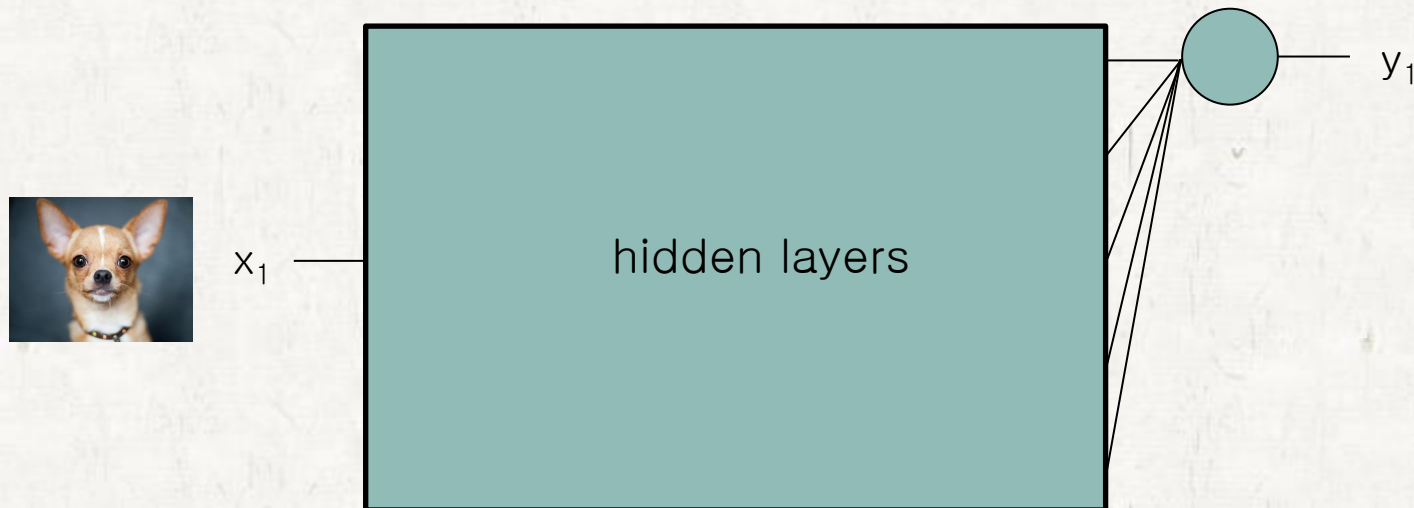
(2470,2460, 2450, 2470, 2480) $\rightarrow$ (2470,2460)



Neural Network Design

- How many inputs? How many outputs?
 - Image Classification

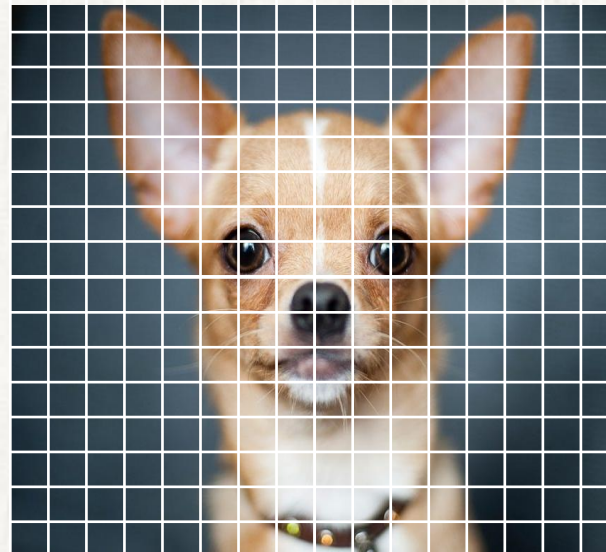
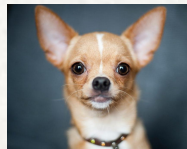
$$\left(\begin{array}{c} \text{Chihuahua Image} \\ , 1 \end{array} \right) \quad \left(\begin{array}{c} \text{Black Cat Image} \\ , 0 \end{array} \right)$$



Neural Network Design

- How many inputs? How many outputs?
 - Image Classification

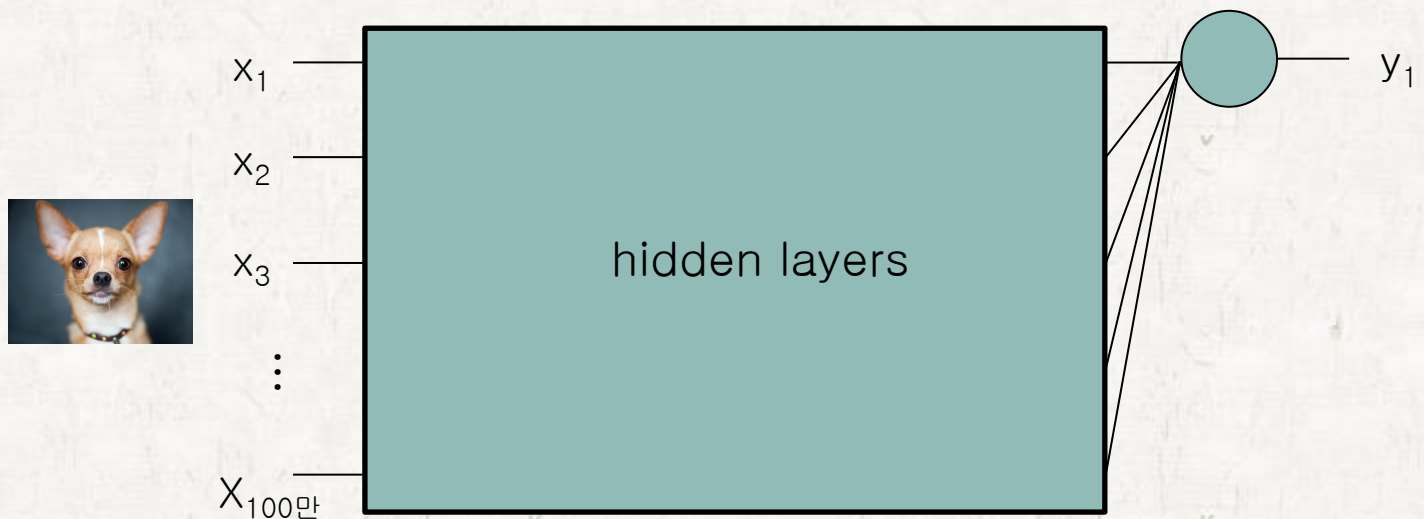
$$\left(\begin{array}{c} \text{Image of a Chihuahua} \\ , \\ 1 \end{array} \right) \quad \left(\begin{array}{c} \text{Image of a black cat} \\ , \\ 0 \end{array} \right)$$



Neural Network Design

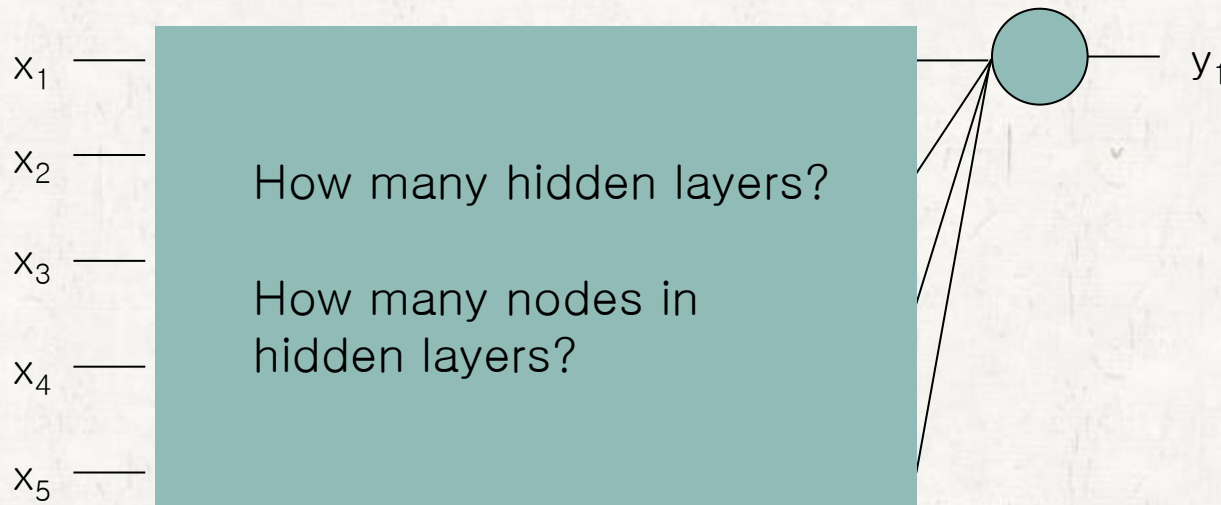
- How many inputs? How many outputs?
 - Image Classification

$$\left(\begin{array}{c} \text{Chihuahua Image} \\ 1 \end{array} \right) \quad \left(\begin{array}{c} \text{Black Cat Image} \\ 0 \end{array} \right)$$



Neural Network Design

- How many hidden layers?
How many nodes in hidden layers?
- 음.. 원하시는데로 하시면 돼요. 그러나
- 쉬운 문제면, layer 조금 node 조금
- 어려운 문제면, layer 많이 node 많이



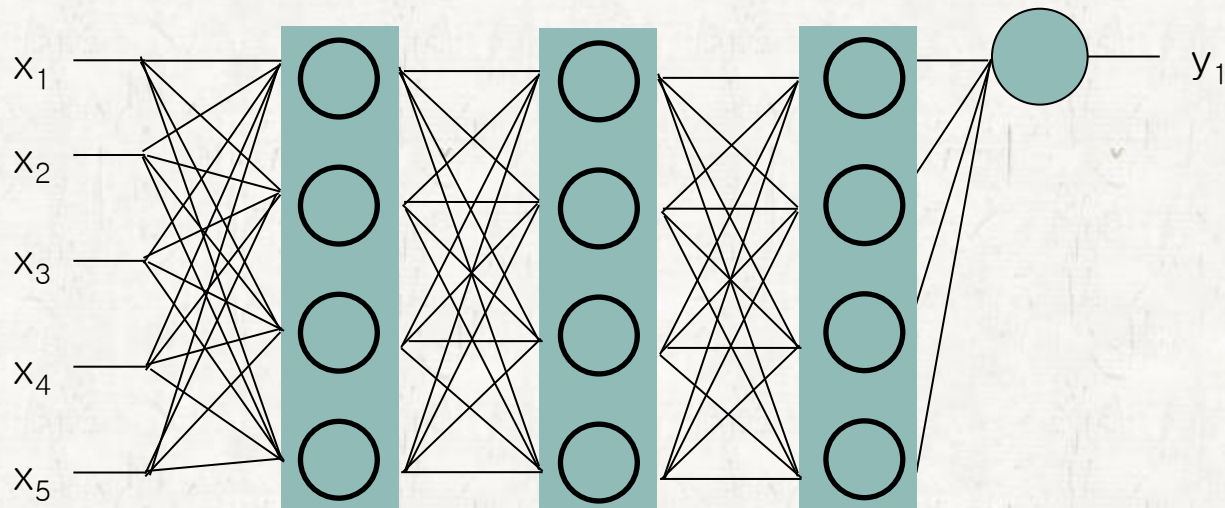
Neural Network Design

- How many hidden layers?
How many nodes in hidden layers?
- Stock Index Prediction

(2500, 2550, 2530, 2540, 2550) $\rightarrow$ 2600

(2400, 2410, 2420, 2430, 2440) $\rightarrow$ 2450

(2470, 2460, 2450, 2470, 2480) $\rightarrow$ 2470



Neural Network Design

- How many hidden layers?

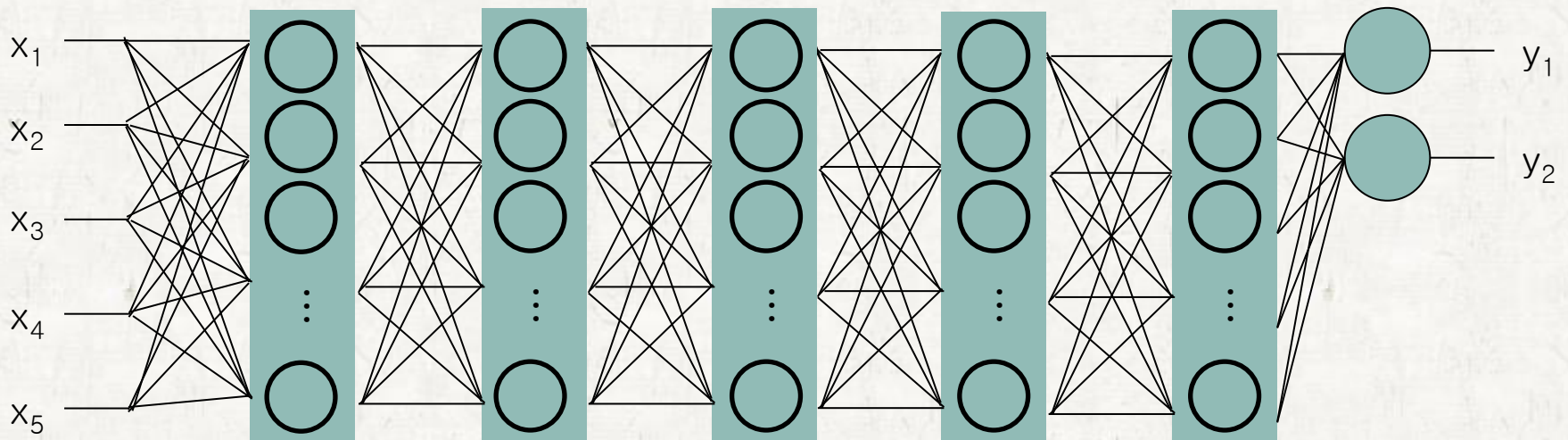
How many nodes in hidden layers?

- Stock Index Prediction

(2500,2550, 2530, 2540, 2550) $\rightarrow$ (2600,2580)

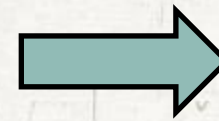
(2400,2410, 2420, 2430, 2440) $\rightarrow$ (2450,2460)

(2470,2460, 2450, 2470, 2480) $\rightarrow$ (2470,2460)



Neural Network Design

- How many hidden layers?
How many nodes in hidden layers?
- Image Classification

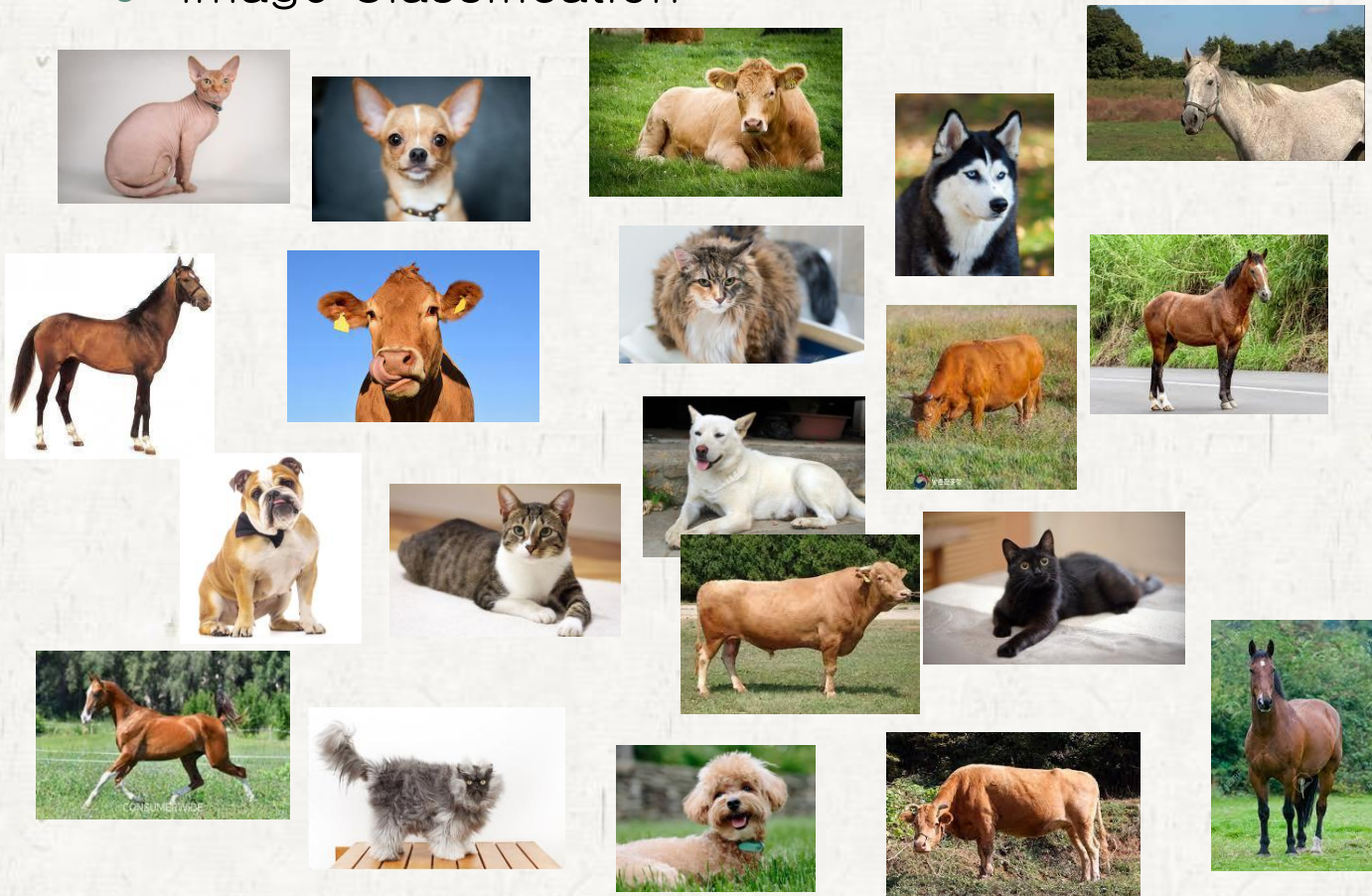


개

고양이

Neural Network Design

- How many hidden layers?
How many nodes in hidden layers?
- Image Classification



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소